

Engine Performance Troubleshooting Tree - CM850 Electronic Control System

Symptoms

- Engine Acceleration or Response Poor
- Cranking Fuel Pressure Low
- Engine Operating Fuel Pressure Low
- Engine Decelerates Slowly
- Engine Difficult to Start or Will **Not** Start (Exhaust Smoke)
- Engine Difficult to Start or Will **Not** Start (No Exhaust Smoke)
- Engine Power Output Low
- Engine Runs Rough at Idle
- Engine Runs Rough or Misfires
- Engine Speed Surges at Low or High Idle
- Engine Speed Surges Under Load or in Operating Range
- Smoke, Black - Excessive
- Smoke, White - Excessive
- Engine Shuts Off or Dies Unexpectedly or Dies During Deceleration
- Engine Starts But Will **Not** Keep Running
- Engine Will **Not** Reach Rated Speed (RPM)
- Intake Manifold Pressure (Boost) Below Normal
- Intake Manifold Pressure (Boost) Above Normal

How To Use This Tree

This symptom tree can be used to troubleshoot all performance-based symptoms listed above. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom. Perform the list of troubleshooting in the sequence shown in the Specifications/Repair section of the tree.

Shoptalk

Driveability is a term that in general describes vehicle performance on the road. Driveability problems for an engine can be caused by several different factors. Some of the factors are engine-related and some are **not**. Before troubleshooting it is important to determine the exact complaint and whether the engine has a real driveability problem, or if it simply does **not** meet driver expectations.

Low power is a term that is used in the field to describe many different performance problems. Low power is defined as the inability of the engine to produce the power necessary to move the vehicle at a speed that can be reasonably expected under the given conditions of load, grade, wind, and so on.

Poor acceleration or response is described as the inability of the vehicle to accelerate satisfactorily from a stop or from the bottom of a grade. It can also be the lag in acceleration during an attempt to pass or overtake another vehicle at conditions less than rated speed and load. Poor acceleration or response is difficult to troubleshoot since it can be caused by several factors.

Included in this engine performance tree is information for troubleshooting engines with the exhaust gas temperature sensor. This information is contained in Step 4.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Perform basic troubleshooting procedures.	
	STEP 1A. Check for active fault codes or high counts of inactive fault codes.	Active fault codes or high counts of inactive fault codes?
	STEP 1B. Perform basic troubleshooting checks.	All steps verified to be correct?
STEP 2.	Determination of engine symptom.	
	STEP 2A. Low power, poor acceleration, or poor response.	Engine symptom - Low Power, Poor Acceleration, or Poor Response?
	STEP 2B. Engine misfire, engine speed surge, or engine speed unstable.	Engine symptom - Engine Misfire, Engine Speed Surge, or Engine Speed Unstable?
	STEP 2C. Excessive white or black smoke.	Engine symptom - Excessive White or Black Smoke?
	STEP 2D. Low intake manifold pressure (boost).	Engine symptom - Low Intake Manifold Pressure (boost)?

STEPS		SPECIFICATIONS
	STEP 2E. High intake manifold pressure (boost).	Engine symptom - High Intake Manifold Pressure (boost)?
	STEP 2F. Engine will not start or difficult to start, engine shuts off unexpectedly.	Engine symptom - Engine Will Not Start or Difficult to Start, Engine Shuts Off Unexpectedly?
STEP 3.	No-start troubleshooting procedures.	
	STEP 3A. Verify the operation of cold weather starting aids.	Necessary cold weather starting aids operating properly?
	STEP 3B. Check for other fault codes that explain a no-start condition.	Any fault codes that can cause a no-start condition become active during cranking?
	STEP 3C. Check engine speed during cranking.	Engine cranking speed greater than 150 rpm?
	STEP 3D. Check the ECM keyswitch voltage.	Keyswitch voltage equal to battery voltage?
	STEP 3E. Check the ECM battery supply voltage.	ECM battery supply voltage equal to the battery voltage?
	STEP 3F. Check fuel rail pressure to the injectors.	Fuel rail pressure greater than 300 bar [4351 psi] while cranking?
	STEP 3G. Check for fuel pressure from the gerotor pump while cranking the engine.	Fuel pressure greater than 500 kPa [73 psi]?
	STEP 3G-1. Check the inlet fuel restriction.	Restriction higher than the specification?
	STEP 3G-2. Check fuel lift pump pressure.	Fuel pressure 3 bar [44 psi] after pump operating for 30 seconds?
	STEP 3G-3. Check if engine starts.	Engine starts?
	STEP 3G-4. Check for proper operation of the check valve near the fuel lift pump.	Check valve near the fuel lift pump operating properly?
	STEP 3G-5. Check if the engine starts.	Engine starts?

STEPS		SPECIFICATIONS
	STEP 3H. Check for external fuel rail (high pressure) leaks.	Fuel rail leakage present?
	STEP 3I. Check injector solenoid operation.	Injector tester tool show a green light?
	STEP 3J. Measure the injector drain flow from all injectors.	Drain fuel flow from the fuel drain block greater than the specification?
	STEP 3J-1. Isolate the injector drain flow from each of the injectors.	Drain fuel flow from the injector greater than the specification?
	STEP 3K. Use INSITE™ electronic service tool to perform a fuel pump actuator click test.	Fuel pump actuator click when commanded with the diagnostic test in INSITE™ electronic service tool?
	STEP 3L. Measure the fuel drain flow from the fuel pressure relief valve.	Drain fuel flow from the fuel pressure relief valve greater than the specification?
STEP 4.	Exhaust gas temperature analysis.	
	STEP 4A. Check for exhaust gas temperature (EGT) sensors.	EGT sensors present?
	STEP 4B. Check for water-cooled exhaust manifold.	Engine has a water-cooled exhaust manifold?
	STEP 4C. Check the cylinder temperatures using an infrared thermometer.	Temperature sensor reading at any cylinder more than 67°C or 120°F different from the average?
	STEP 4D. Check exhaust gas temperature (EGT) readings.	Any of the exhaust gas temperature sensors read: For Thermocouple: 474°C [885°F]? For Thermistors: 600°C [1112°F]?
	STEP 4D-1. Check the engine harness and the ECM 1 and ECM 2 connectors for affected cylinders.	Dirty or damaged pins?
	STEP 4D-2. Check the supply voltage at the affected ECM connector.	4.75 to 5.25-VDC?

STEPS		SPECIFICATIONS
	STEP 4D-3. Check the exhaust gas temperature sensor converter and engine harness connector pins.	Dirty or damaged pins?
	STEP 4D-4. Check for an open circuit in the exhaust gas temperature sensor or sensor converter supply.	Less than 10 ohms?
	STEP 4D-5. Check for a short circuit to ground.	Greater than 100k ohms?
	STEP 4E. Check for significant temperature difference.	Temperature sensor reading at any cylinder more than 45°C or 81°F different from the average?
	STEP 4E-1. Check for shared turbochargers/air handling passages.	Are cylinders that share a particular turbocharger affected?
STEP 5.	Fuel system troubleshooting procedures.	
	STEP 5A. Check for fault codes.	Fuel system fault codes active?
	STEP 5B. Measure the fuel inlet restriction.	Fuel restriction above specifications?
	STEP 5C. Check stage one fuel filter head check valves for proper operation.	Fuel filter head check valves in good condition and the passageway free of restrictions?
	STEP 5D. Check injector solenoid operation.	Injector tester tool show a green light?
	STEP 5E. Use INSITE™ electronic service tool to perform fuel pump actuator click test.	Fuel pump actuator click when commanded with the diagnostic test in INSITE™ electronic service tool?
	STEP 5F. Measure the injector drain flow from all injectors.	Drain fuel flow from the fuel drain block greater than the specification?
	STEP 5F-1. Isolate the injector drain flow from each of the injectors.	Drain fuel flow from the injector greater than the specification?
	STEP 5G. Perform single cylinder cut-out test.	Miss or excessive smoke be attributed to a single cylinder?

STEPS		SPECIFICATIONS
	STEP 5H. Audio check for injector operation.	Can a miss be heard from any individual injector?
STEP 6.	Air handling troubleshooting procedures.	
	STEP 6A. Inspect the turbocharger blades for damage.	Damage found on turbocharger blades?
	STEP 6B. Check the turbocharger axial and radial clearances.	Turbocharger axial and radial bearing clearance within specification?
	STEP 6C. Inspect the air cooling system.	Charge-air cooler free of cracks or other damage?
	STEP 6D. Check air intake restrictions.	Intake restriction greater than 635 mm-H ₂ O [25 in-H ₂ O]?
STEP 7.	Electronic feature troubleshooting procedures.	
	STEP 7A. Verify throttle pedal travel.	Throttle position read 0 percent when the throttle is released and 100 percent when the throttle is depressed?
	STEP 7B. Check barometric pressure sensor accuracy.	INSITE™ electronic service tool reading is within 102 mm-Hg [4 in-Hg] of local barometric pressure?
	STEP 7C. Check intake manifold pressure sensor accuracy.	Intake manifold pressure reading is less than 102 mm-Hg [4 in-Hg]?
STEP 8.	Base engine troubleshooting procedures.	
	STEP 8A. Verify overhead adjustments are correct.	Overhead settings within the reset limits?
	STEP 8B. Check exhaust restriction.	Exhaust back pressure less than 75 mm-Hg [3 in-Hg]?

STEP 1. Perform basic troubleshooting procedures.

STEP 1A. Check for active fault codes or high counts of inactive fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for active fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Active fault codes or high counts of inactive fault codes? YES Repair : Follow the electronic fault code trees for troubleshooting procedures.	Repair complete
	Active fault codes or high counts of inactive fault codes? NO	1B

STEP 1B. Perform basic troubleshooting checks.**Conditions :**

None.

Action	Specification/Repair	Next Step
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The following items must be checked or verified before continuing: • Verify fuel level in the tanks • Verify there have not been any changes to CPL components on the engine • Verify fuel grade is correct for application • Verify engine is operating within the recommended altitude • Verify engine oil is at the correct level • Verify engine parasitics have not changed • Verify engine duty cycle has not changed • Verify engine cranking speed is greater than 150 rpm • Verify battery voltage is within specification.	All steps have been verified to be correct? YES	2A
	All steps have been verified to be correct? NO Repair : Correct or repair and verify complaint is no longer present after repair.	Repair complete

STEP 2. Determination of engine symptoms.

STEP 2A. Low power, poor acceleration, or poor response.

Conditions :

None.

Action	Specification/Repair	Next Step

Interview the driver and verify the complaint. • Interview the operator.	Engine symptom - Low Power, Poor Acceleration, or Poor Response? YES Repair : Perform the troubleshooting steps in the recommended order listed below: • Step 4 - Exhaust Gas Temperature Analysis EGT only • Step 5 - Fuel System Checks • Step 6 - Air Handling Checks • Step 7 - Electronics Checks • Step 8 - Base Engine Checks.	Perform the troubleshooting steps suggested in the repair procedure.
	Engine symptom - Low Power, Poor Acceleration, or Poor Response? NO	2B

STEP 2B. Engine misfire, engine speed surge, or engine speed unstable.

Conditions : None.		
Action	Specification/Repair	Next Step

Interview the driver and verify the complaint. • Interview the operator.	Engine symptom - Engine Misfire, Engine Speed Surge, or Engine Speed Unstable? YES Repair : Perform the troubleshooting steps in the recommended order listed below: • Step 4 - Exhaust Gas Temperature Analysis EGT only • Step 5 - Fuel System Checks • Step 6 - Air Handling Checks • Step 7 - Electronics Checks • Step 8 - Base Engine Checks.	Perform the troubleshooting steps suggested in the repair procedure.
	Engine symptom - Engine Misfire, Engine Speed Surge, or Engine Speed Unstable? NO	2C

STEP 2C. Excessive white or black smoke.

Conditions : None.		
Action	Specification/Repair	Next Step

Interview the driver and verify the complaint. • Interview the operator.	Engine symptom - Excessive White or Black Smoke? YES Repair : Perform the troubleshooting steps in the recommended order listed below: • Step 4 - Exhaust Gas Temperature Analysis EGT only • Step 5 - Fuel System Checks • Step 6 - Air Handling Checks • Step 8 - Base Engine Checks.	Perform the troubleshooting steps suggested in the repair procedure.
	Engine symptom - Excessive White or Black Smoke? NO	2D

STEP 2D. Low intake manifold pressure (boost).

Conditions : None.		
Action	Specification/Repair	Next Step

Interview the driver and verify the complaint. Interview the operator.	Engine symptom - Low Intake Manifold Pressure (boost)? YES Repair : Perform the troubleshooting steps in the recommended order listed below: - Step 6 - Air Handling Checks - Step 5 - Fuel System Checks - Step 8 - Base Engine Checks.	Perform the troubleshooting steps suggested in the repair procedure.
	Engine symptom - Low Intake Manifold Pressure (boost)? NO	2E

STEP 2E. High intake manifold pressure (Boost).

Conditions : None		
Action	Specification/Repair	Next Step

Interview the operator and verify the compliant. • Interview the operator.	Engine symptom - High Intake Manifold Pressure (boost)? YES Repair : Perform the troubleshooting steps in the recommended order listed below: • Step 7 - Electronics Checks • Step 5 - Fuel System Checks.	Perform the troubleshooting steps suggested in the repair procedure.
	Engine symptom - High Intake Manifold Pressure (boost)? NO	2F

STEP 2F. Engine will not start or difficult to start, engine shuts off unexpectedly.

Conditions : None.		
Action	Specification/Repair	Next Step

Interview the driver and verify the complaint. • Interview the operator.	Engine symptom - Engine Will Not Start or Difficult to Start, Engine Shuts Off Unexpectedly? YES Repair : Perform the troubleshooting steps in the recommended order listed below: • Step 3 - No Start Checks • Step 5 - Fuel System Checks • Step 6 - Air Handling Checks • Step 7 - Electronics Checks.	Perform the troubleshooting steps suggested in the repair procedure.
	Engine symptom - Engine Will Not Start or Difficult to Start, Engine Shuts Off Unexpectedly? NO	Return to correct symptom tree

STEP 3. No-start troubleshooting procedures.

STEP 3A. Verify the operation of cold weather starting aids.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step

Check cold weather starting aids. Reference the Operation of Diesel Engines in Cold Climates, Bulletin 3379009.	Necessary cold weather starting aids operating properly? YES	3B
	Necessary cold weather starting aids operating properly? NO Repair : Install or repair cold weather starting aids. Reference the Operation of Diesel Engines in Cold Climates, Bulletin 3379009.	Repair complete

STEP 3B. Check for other fault codes that explain a no-start condition.

Conditions : - Turn keyswitch ON. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step
Use INSITE™ electronic service tool to read fault code information. Check for fault codes that come active during a unsuccessful start attempt that can be the cause for a no-start condition.	Any fault codes that can cause a no-start condition become active during cranking? YES Repair : Follow the electronic fault code trees for the appropriate troubleshooting procedures.	Repair complete
	Any fault codes that can cause a no-start condition become active during cranking? NO	3C

STEP 3C. Check engine speed during cranking.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Use INSITE™ electronic service tool to monitor engine speed and fuel rail pressure while cranking the engine. • Check engine speed during cranking. • Monitor fuel rail pressure during cranking, (300 bar [4351 psi] minimum pressure is required to start the engine). If lower fuel rail pressure is observed after completing all sub-steps of Step 3, proceed to Step 5 for further fuel system troubleshooting procedures.	Engine cranking speed greater than 150 rpm? YES	3D
	Engine cranking speed greater than 150 rpm? NO Repair : Find and correct the cause for low cranking speed. Check the batteries, engine starting motor, and accessory loads. See the Engine Will Not Crank or Cranks Slowly troubleshooting symptom tree.	Repair complete

STEP 3D. Check the ECM keyswitch voltage.

Conditions :

- Disconnect the engine harness from the ECM 50-pin connector.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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Measure the signal voltage from the keyswitch input SIGNAL wire of the engine harness to the engine block ground. • Measure the keyswitch voltage with the keyswitch in the ON position and also with the keyswitch in the cranking position. Refer to the circuit diagram or wiring diagram for connector pin identification.	Keyswitch voltage equal to battery voltage? YES	3E
	Keyswitch voltage equal to battery voltage? NO Repair : • Repair or replace the OEM power harness. See equipment manufacturer service information. • Repair or replace the keyswitch. Refer to Procedure 019-064 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-064.html) • Check the battery connections and fuse terminals. See equipment manufacturer service information.	Repair complete

STEP 3E. Check the ECM battery supply voltage.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ECM power harness from the ECM.

Action	Specification/Repair	Next Step

Measure the voltage from the ECM battery SUPPLY (-) pin to the ECM battery SUPPLY (+) pins in the ECM power harness connector. • Measure the ECM voltage with the keyswitch in the ON position and also with the keyswitch in the cranking position. • Refer to the wiring diagram for connector pin identification.	ECM battery supply voltage equal to the battery voltage? YES	3F
	ECM battery supply voltage equal to the battery voltage? NO Repair : See equipment manufacturer service information. Repair or replace the ECM power harness. • Check the battery connections and fuse terminals. See equipment manufacturer service information.	Repair complete

STEP 3F. Check fuel rail pressure to the injectors.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Measure the fuel rail pressure output pressure while cranking the engine. The cranking speed must be greater than 150rpm for 30 seconds.	Fuel rail pressure greater than 300 bar [4351 psi] while cranking? YES	5B
	Fuel rail pressure greater than 300 bar [4351 psi] while cranking? NO Repair : Repair or replace the components associated with the leaks in the fuel system. Install the fuel check ball valve in the correct direction if it was found to be installed incorrectly.	3G

STEP 3G. Check for fuel pressure from the gerotor pump while cranking the engine.

Conditions : - Turn keyswitch ON. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step
Attempt to start the engine by engaging the engine starting motor for at least 30 continuous seconds. Use INSITE™ electronic service tool to monitor fuel pressure.	Fuel pressure greater than 500 kPa [73 psi]? YES	3H
	Fuel pressure greater than 500 kPa [73 psi]? NO	3G-1

STEP 3G-1. Check the inlet fuel restriction.

Conditions : Turn keyswitch ON.		
Action	Specification/Repair	Next Step

	Restriction higher than the specification? YES	Repair complete
	Restriction higher than the specification? NO Repair : • If the inlet restriction is higher than specifications, refer to the OEM service manual to determine the source of the high restriction. • If the stage 1 filter restriction is higher than specifications, replace the stage one filter. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-020 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-020-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-020 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-020-tr.html)	3G-2

STEP 3G-2. Check fuel lift pump pressure.

Conditions :

- Turn keyswitch ON.
- Attach a fuel pressure gauge to the fuel filter head stage one filter outlet.

Action	Specification/Repair	Next Step
Measure the fuel lift pump output pressure. • Turn keyswitch ON, but engine not operating. Listen for the fuel lift pump to operate. The pump will run for 30 seconds then shut off.	Fuel pressure 3 bar [44 psi] after pump operating for 30seconds? YES Repair : • Replace the air bleed check valve. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-021 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-021-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-021 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-021-tr.html)	3G-3
	Fuel pressure 3 bar [44 psi] after pump operating for 30seconds? NO	3G-4

STEP 3G-3. Check if engine starts.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step
Check if engine starts. • Attempt to start the engine.	Engine starts? YES Repair : The replacement of the air bleed check valve in the previous step corrected the problem.	Repair complete
	Engine starts? NO Repair : • Replace the the fuel pump.. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)	Repair complete

STEP 3G-4. Check for proper operation of the check valve near the fuel lift pump.

Conditions : • Turn keyswitch ON.		
Action	Specification/Repair	Next Step

Inspect the check valve for proper operation. • Turn keyswitch ON, but engine not operating.	Check valve near the fuel lift pump operating correctly? YES Repair : • Replace the air bleed check valve. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-021 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-021-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-021 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-021-tr.html)	3G-5
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	Check valve near the fuel lift pump operating correctly? NO Repair : • Replace the check valve near the fuel lift pump. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-008 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-008.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-008 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-008.html)	Repair complete
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STEP 3G-5. Check if engine starts.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step

Check if engine starts. Attempt to start the engine.	Engine starts? YES Repair : The replacement of the air bleed check valve in the previous step corrected the problem.	Repair complete
	Engine starts? NO Repair : • Replace the check valve near the fuel lift pump. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 005-045 in Section 5. (/qs3/pubsys2/xml/en/procedures/28/28-005-045-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 005-045 in Section 5. (/qs3/pubsys2/xml/en/procedures/56/56-005-045-tr.html)	Repair complete

STEP 3H. Check for external fuel rail (high pressure) leaks.

Conditions :

Crank the engine to develop fuel pressure.

Action	Specification/Repair	Next Step
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Inspect the fuel rail for leaks. - Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-051-tr.html)	Fuel rail leakage present? YES Repair : Repair fuel rail or leaking component.	Repair complete
	Fuel rail leakage present? NO Repair : Connect the fuel supply lines in the correct port.	3I

STEP 3I. Check injector solenoid operation.

Conditions :

- Turn keyswitch ON
- Attach the Injector Tester, Part Number 2892293, to an injector.

Action	Specification/Repair	Next Step

Test all of the injectors using the injector tester tool. • Turn keyswitch ON. • Attach the injector tester tool to the two pin injector connector. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Injector tester tool show a green light? YES	3J
	Injector tester tool show a green light? NO Repair : Replace any malfunctioning injectors. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Repair complete

STEP 3J. Measure the injector drain flow from all injectors.

Conditions :

- Crank engine to at least 150 rpm.
- Unplug all injectors.
- Connect appropriate service tools to measure injector drain flow at the fuel drain block.

Action	Specification/Repair	Next Step
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Measure the injector return fuel drain flow at the fuel drain block. - Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Drain fuel flow from the fuel drain block greater than the specification? YES	3J-1
	Drain fuel flow from the fuel drain block greater than the specification? NO	3K

STEP 3J-1. Isolate the injector drain flow from each of the injectors.

Conditions :

- Crank engine to at least 150 rpm.
- Unplug all injectors.
- Connect appropriate service tools to measure injector drain flow at each cylinder head.

Action	Specification/Repair	Next Step
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Measure the injector return fuel drain flow at the fuel drain block. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Drain fuel flow from the injector greater than the specification? YES Repair : Replace any malfunctioning injectors. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Repair complete
	Drain fuel flow from the injector greater than the specification? NO	3K

STEP 3K. Use INSITE™ electronic service tool to perform the fuel pump actuator click test.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool..

Action	Specification/Repair	Next Step
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Perform the fuel pump actuator click test • Use INSITE™ electronic service tool to perform the fuel pump actuator click test.	Fuel pump actuator click when commanded with the diagnostic test in INSITE™ electronic service tool? YES	3L
	Fuel pump actuator click when commanded with the diagnostic test in INSITE™ electronic service tool? NO Repair : Replace the fuel pump actuator. • Refer to Procedure 019-117 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-117.html)	Repair complete

STEP 3L. Measure the fuel drain flow from the fuel pressure relief valve.

Conditions :

- Crank engine to at least 150 rpm.
- Unplug all injectors.
- Connect appropriate service tools to measure fuel drain flow from the fuel pressure relief valve.

Action	Specification/Repair	Next Step

Measure the injector return fuel drain flow at the fuel drain block. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-061 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-061 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html)	Drain fuel flow from the fuel pressure relief valve greater than the specification? YES Repair : Replace the fuel pressure relief valve. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-061 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-061 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html)	Repair complete
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	Drain fuel flow from the fuel pressure relief valve greater than the specification? NO Repair : • Replace the the fuel pump. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)	Repair complete
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STEP 4. Exhaust gas temperature analysis.

STEP 4A. Check for exhaust gas temperature sensors.

Conditions :

- Check for exhaust gas temperature sensors.
- Refer to Procedure 019-013 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-013.html)

Action	Specification/Repair	Next Step
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Check for exhaust gas temperature sensors. Refer to Procedure 019-013 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-013.html)	EGT sensors present? YES	4D
	EGT sensors present? NO	4B

STEP 4B. Check for water-cooled exhaust manifold.

Conditions :

- Check for water-cooled exhaust manifold.

Action	Specification/Repair	Next Step
Check for water-cooled exhaust manifold. A water-cooled exhaust manifold is a double-wall or triple-wall liquid-cooled exhaust manifold. These are typically used on marine engines.	Engine has a water-cooled exhaust manifold? YES	5A
	Engine has a water-cooled exhaust manifold? NO	4C

STEP 4C. Check cylinder temperatures using an infrared thermometer.

Conditions :

- Turn keyswitch ON.
- Operate the engine at at least 50 percent load for 5 minutes.

Action	Specification/Repair	Next Step
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Use an infrared thermometer to check cylinder temperatures. Manually calculate the average of the temperature readings. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Temperature sensor reading at any cylinder more than 67°C or 120°F different from the average? YES Repair : Replace any malfunctioning injectors. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Repair complete
	Temperature sensor reading at any cylinder more than 67°C or 120°F different from the average? NO	5A

STEP 4D. Check exhaust gas temperature (EGT) readings.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check the exhaust gas temperature (EGT) sensor readings. • Use INSITE™ electronic service tool to take a datalog of all exhaust gas temperature sensors. • Thermocouples have a 3 pin connector or a converter box • Thermistors have 2 pin individual connectors and no converter boxes.	Any of the exhaust gas temperature sensors read: • For Thermocouple: 474°C [885°F]? • For Thermistors: 600°C [1112°F]? YES	4D-1
	Any of the exhaust gas temperature sensors read: • For Thermocouple: 474°C [885°F]? • For Thermistors: 600°C [1112°F]? NO	4E

STEP 4D-1. Check the engine harness and the ECM 1 and ECM 2 connectors for affected cylinders.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the affected ECM connector (Cylinders 1,3,5,7: ECM 1 60-pin connector; Cylinders 2,4,6,8: ECM 2 60-pin connector; Cylinders 9,11,13,15: ECM 1 50-pin connector; Cylinders 10,12,14,16: ECM 2 50-pin connector).

Action	Specification/Repair	Next Step
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Inspect the engine harness and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the engine harness connector or ECM connector. • Flush the dirt, debris, or moisture from the connector pins. Use electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Replace the damaged section of the engine harness. • Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-043.html) • Call for authorization to replace the ECM. Upon receipt of authorization, replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)	4D-5
	Dirty or damaged pins? NO	4D-2

STEP 4D-2. Check the supply voltage at the affected ECM connector.

Conditions :

- (This step is for engines with thermocouples **only**. Engines with thermocouple-type sensors have either convertor boxes or 3-pin EGT connectors. Go to Step 4D-3 for engines with thermistors.)
- Turn keyswitch OFF.
- Disconnect the engine harness connector from the affected ECM connector (Cylinders 1,3,5,7: ECM 1 60-pin connector; Cylinders 2,4,6,8: ECM 2 60-pin connector; Cylinders 9,11,13,15: ECM 1 50-pin connector; Cylinders 10,12,14,16: ECM 2 50-pin connector).
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
Check the supply voltage at the ECM connector. • Measure the voltage from the exhaust gas temperature sensor converter 5 volt SUPPLY pin of the ECM connector to the exhaust gas temperature sensor converter RETURN pin of the ECM connector. Refer to the wiring diagram for connector pin identification.	4.75 to 5.25-VDC? YES	4D-3
	4.75 to 5.25-VDC? NO Repair : Call for authorization to replace the ECM. Upon receipt of authorization, replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)	4D-5

STEP 4D-3. Check the exhaust gas temperature sensor converter and engine harness connector pins.**Conditions :**

- Turn keyswitch OFF.
- Disconnect the engine harness connector and all EGT interconnects from the exhaust gas temperature sensors.

Action	Specification/Repair	Next Step
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Inspect the engine harness connector and exhaust gas temperature sensor connector pins or converter connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the engine harness connector or exhaust gas temperature sensor converter connector. • Flush the dirt, debris, or moisture from the connector pins. Use electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Replace the damaged section of the engine harness. • Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-043.html) • Replace the exhaust gas temperature sensor converter. • Refer to Procedure 019-450 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-450.html)	4D-5
	Dirty or damaged pins? NO	4D-4

STEP 4D-4. Check for an open circuit in the exhaust gas temperature sensor or sensor converter supply.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the affected exhaust gas temperature sensor or sensor converter.
- Disconnect the engine harness connector from the affected ECM connector (Cylinders 1,3,5,7: ECM 1 60-pin connector; Cylinders 2,4,6,8: ECM 2 60-pin connector; Cylinders 9,11,13,15: ECM 1 50-pin connector; Cylinders 10,12,14,16: ECM 2 50-pin connector).

Action	Specification/Repair	Next Step
Check for an open circuit. • Measure the resistance from the exhaust gas temperature sensor converter 5 volt SUPPLY pin of the engine harness ECM connector to the exhaust gas temperature sensor converter SUPPLY pin of the affected exhaust gas temperature sensor converter connector. Refer to the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms? YES	4D-5
	Less than 10 ohms? NO Repair : • Replace the exhaust gas temperature sensor. • Refer to Procedure 019-013 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-013.html) • Replace the exhaust gas temperature sensor converter. • Refer to Procedure 019-450 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-450.html)	4E

STEP 4D-5. Check for a short circuit to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the affected exhaust gas temperature sensor or sensor converter.
- Disconnect the engine harness connector from the affected ECM connector (Cylinders 1,3,5,7: ECM 1 60-pin connector; Cylinders 2,4,6,8: ECM 2 60-pin connector; Cylinders 9,11,13,15: ECM 1 50-pin connector; Cylinders 10,12,14,16: ECM 2 50-pin connector).

Action	Specification/Repair	Next Step
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Check for a short circuit to ground. • Check the exhaust gas temperature sensor or sensor convertor for short circuit to ground.	Greater than 100k ohms? YES Repair : The EGT system has suffered an in-range malfunction. • Replace the exhaust gas temperature sensor. • Refer to Procedure 019-013 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-013.html) • Replace the exhaust gas temperature sensor converter. • Refer to Procedure 019-450 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-450.html)	4E
	Greater than 100k ohms? NO Repair : • Troubleshoot all harnesses connected in series to determine which contains the short circuit to ground. • Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-043.html)	4E

STEP 4E. Check for significant temperature difference.

Conditions :

- Turn keyswitch ON and operate the engine to at least 50 percent load for 5 minutes.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check the exhaust gas temperature (EGT) sensor readings. • Use INSITE™ electronic service tool to take a datalog of all exhaust gas temperature sensors and take the average using INSITE™ electronic service tool Average Exhaust Temperature Parameter.	Temperature sensor reading at any cylinder more than 45°C or 81°F different from the average? YES	4E-1
	Temperature sensor reading at any cylinder more than 45°C or 81°F different from the average? NO	5A

STEP 4E-1. Check for shared turbochargers/air handling passages.**Conditions :**

- Check the air handling layout for common turbochargers or passages for particular cylinders.

Action	Specification/Repair	Next Step
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	Are cylinders that share a particular turbocharger affected? YES	6A
	Are cylinders that share a particular turbocharger affected? NO Repair : Replace the injectors. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Repair complete

STEP 5. Fuel system troubleshooting procedures.

STEP 5A. Check for fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for fault codes. • Use INSITE™ electronic service tool to read the fault code information. • Determine if there are active fuel system fault codes related to the complaint.	Fuel system fault codes active? YES Repair : Follow the electronic fault code trees for the appropriate troubleshooting procedures.	Repair complete
	Fuel system fault codes active? NO	5B

STEP 5B. Measure the fuel inlet restriction.**Conditions :**

Measure the fuel inlet restriction.

- Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-020 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-020-tr.html>)
- Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-020 in Section 6. (</qs3/pubsys2/xml/en/procedures/56/56-006-020-tr.html>)

Action	Specification/Repair	Next Step
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Check for fuel inlet restriction. - Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-020 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-020-tr.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-020 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-020-tr.html)	Fuel restriction above specifications? YES Repair : Find the correct cause of the high inlet restriction. Look for plugged OEM fuel filters or screens, a restricted lift pump bypass check valve, pinched OEM fuel lines, or a restricted stand pipe in the OEM fuel tank.	Repair complete
	Fuel restriction above specifications? NO	5C

STEP 5C. Check stage one fuel filter head check valves for proper operation.

Conditions :

- Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-008 in Section 6. (</qs3/pubsys2/xml/en/procedures/102/102-006-008.html>)
- Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-008 in Section 6. (</qs3/pubsys2/xml/en/procedures/102/102-006-008.html>)

Action	Specification/Repair	Next Step
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Check fuel filter head check valves for proper operation. • The check valve prevents back-flow of fuel through the first stage filter during priming and operation, when installed properly. • A stuck check valve will not allow fuel flow from the lift pump to prime the engine.	Fuel filter head check valves in good condition and the passageway free of restrictions? YES	5D
	Fuel filter head check valves in good condition and the passageway free of restrictions? NO Repair : • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-008 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-008.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-008 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-008.html)	Repair complete

STEP 5D. Check injector solenoid operation.

Conditions :

- Turn keyswitch ON.
- Attach the Injector Tester, Part Number 2892293, to an injector.

Action	Specification/Repair	Next Step
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Test all of the injectors using the injector tester tool. • Turn the keyswitch ON, and attach the injector tester tool to the 2 pin injector connector. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Injector tester tool show a green light? YES	5E
	Injector tester tool show a green light? NO Repair : Replace any malfunctioning injectors. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Repair complete

STEP 5E. Use INSITE™ electronic service tool to perform fuel pump actuator click test.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Perform the fuel pump actuator click test. • Use INSITE™ electronic service tool to perform fuel pump actuator click test.	Fuel pump actuator click when commanded with the diagnostic test in INSITE™ electronic service tool? YES	5F
	Fuel pump actuator click when commanded with the diagnostic test in INSITE™ electronic service tool? NO Repair : Replace the fuel pump actuator. • Refer to Procedure 019-117 in Section 19 (/qs3/pubsys2/xml/en/procedures/05/05-019-117.html)	Repair complete

STEP 5F. Measure the injector drain flow from all injectors.

Conditions :

- Crank engine to at least 150 rpm.
- Unplug all injectors.
- Connect appropriate service tools to measure injector drain flow at the fuel drain block.

Action	Specification/Repair	Next Step

Measure the injector return fuel drain flow at the fuel drain block. - Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Drain fuel flow from the fuel drain block greater than the specification? YES	5F-1
	Drain fuel flow from the fuel drain block greater than the specification? NO	5G

STEP 5F-1. Isolate the injector drain flow from each of the injectors.

Conditions :

- Crank engine to at least 150 rpm.
- Unplug all injectors.
- Connect appropriate service tools to measure injector drain flow at each cylinder head.

Action	Specification/Repair	Next Step
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Measure the injector return fuel drain flow at the cylinder head. - Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Drain fuel flow from the injector greater than the specification? YES Repair : Replace any malfunctioning injectors. - Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Repair complete
	Drain fuel flow from the injector greater than the specification? NO	5G

STEP 5G. Perform single cylinder cut-out test

Conditions :

- Turn keyswitch ON.
- Operate engine at low idle.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Perform single cylinder cut-out test. • Operate the engine at load. • Use INSITE™ electronic service tool to perform the cylinder cut-out test to disable individual injectors.	Miss or excessive smoke be attributed to a single cylinder? YES Repair : Look for a cause of the complaint, including valve lash and excessive crankcase pressure that can indicate power cylinder damage or camshaft lobe wear. If no other damage is found, replace the fuel injector in the cylinder that was identified using the single cylinder cut-out test. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Repair complete
	Miss or excessive smoke be attributed to a single cylinder? NO	5H

STEP 5H. Audio check for injector operation.

Conditions :

- Turn keyswitch ON.
- Operate engine at low idle.

Action	Specification/Repair	Next Step
Perform an audio check on each injector to verify proper operation. • With the engine running at idle, use a mechanic's stethoscope (Snap-On™, Part Number GA111D, or similar) to identify any individual injector improper operation.	Can a miss be heard from any individual injector? YES Repair : Replace any faulty injectors. • Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	Repair complete
	Can a miss be heard from any individual injector? NO	2A

STEP 6. Air handling troubleshooting procedures.

STEP 6A. Inspect the turbocharger blades for damage.

Conditions :

- Turn keyswitch OFF.
- Remove the intake and exhaust pipes from the turbocharger.

Action	Specification/Repair	Next Step
Inspect the compressor and turbine blades for damage or wear. • Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-033-tr.html)	Damage found on turbocharger blades? YES Repair : Replace the turbocharger assembly. • Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-033-tr.html)	Repair complete
	Damage found on turbocharger blades? NO	6B

STEP 6B. Check the turbocharger axial and radial clearances.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check the turbocharger for correct axial and radial clearance. - Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-033-tr.html)	Turbocharger axial and radial bearing clearances within specification? YES	6C
	Turbocharger axial and radial bearing clearances within specification? NO Repair : Replace the turbocharger assembly. - Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-033-tr.html)	Repair complete

STEP 6C. Inspect the air cooling system.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
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Inspect the charge-air cooler. • Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528, for inspection procedures (QSK60 II stage only). Refer to Procedure 010-003 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-003-tr.html) • Use the following procedures in the QSK45 and QSK60 Service Manual, Bulletin 4021530, for inspection procedures. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html) • Refer to Procedure 010-083 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-083-tr.html)	Charge-air cooler free of cracks or other damage? YES	6D
	Charge-air cooler free of cracks or other damage? NO Repair : Repair or replace the aftercooler and intercooler • Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528 (QSK60 II stage only). Refer to Procedure 010-003 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-003-tr.html) • Use the following procedures in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html) • Refer to Procedure 010-083 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-083-tr.html)	Repair complete

STEP 6D. Check air intake restriction.

Conditions :

- Turn keyswitch OFF.
- Install a vacuum gauge or water manometer in the air intake piping.

Action	Specification/Repair	Next Step
Check air intake system restriction. Install a vacuum gauge into the air intake system. • Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 010-031 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-031-tr.html) • Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 010-031 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-031.html)	Air intake restriction greater than 635 mm-H₂O [25 in-H₂O]? YES	Repair complete
	Air intake restriction greater than 635 mm-H₂O [25 in-H₂O]? NO	2A

STEP 7. Electronic feature troubleshooting procedures.

STEP 7A. Verify throttle pedal travel.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Verify throttle pedal travel. • Use INSITE™ electronic service tool to monitor the throttle position while fully depressing and releasing the throttle pedal.	Throttle position read 0 percent when the throttle is released and 100 percent when the throttle is depressed? YES	7B
	Throttle position read 0 percent when the throttle is released and 100 percent when the throttle is depressed? NO Repair : Determine and correct the cause of the throttle pedal restriction.	Repair complete

STEP 7B. Check barometric pressure sensor accuracy.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Use INSITE™ electronic service tool to monitor the barometric air pressure. Compare INSITE™ electronic service tool barometric pressure readings with the local barometric pressure.	INSITE™ electronic service tool reading is within 102 mm-Hg [4 in-Hg] of local barometric pressure? YES	7C
	INSITE™ electronic service tool reading is within 102 mm-Hg [4 in-Hg] of local barometric pressure? NO Repair : Replace the barometric pressure sensor. Refer to Procedure 019-004 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-004.html)	Repair complete

STEP 7C. Check intake manifold pressure sensor accuracy.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Check intake manifold pressure sensor accuracy. • Use INSITE™ electronic service tool to monitor the value of intake manifold pressure without operating the engine.	Intake manifold pressure reading is less than 102 mm-Hg [4 in-Hg]? YES	8A
	Intake manifold pressure reading is less than 102 mm-Hg [4 in-Hg]? NO Repair : Replace the intake manifold pressure sensor. Refer to Procedure 019-061 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-061.html)	Repair complete

STEP 8. Base engine troubleshooting procedures.

STEP 8A. Verify overhead adjustments are correct.

Conditions :

- Turn keyswitch OFF.

Remove the rocker lever cover.

- Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
- Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html)

Action	Specification/Repair	Next Step

Measure the overhead setting.	Overhead settings within the reset limits? YES	8B
	Overhead settings within the reset limits? NO Repair : Adjust the overhead settings. - Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 003-006 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 003-006 in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html)	Repair complete

STEP 8B. Check exhaust restriction.

Conditions :

- Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 011-009 in Section 11. (/qs3/pubsys2/xml/en/procedures/28/28-011-009.html)
- Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 011-009 in Section 11. (/qs3/pubsys2/xml/en/procedures/56/56-011-009.html)

Action	Specification/Repair	Next Step
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Check the exhaust system back pressure. - Use the following procedure in the K38, K50 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 011-009 in Section 11. (/qs3/pubsys2/xml/en/procedures/28/28-011-009.html) - Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 011-009 in Section 11. (/qs3/pubsys2/xml/en/procedures/56/56-011-009.html)	Exhaust back pressure less than 75 mm-Hg [3 in-Hg]? YES	2A
	Exhaust back pressure less than 75 mm-Hg [3 in-Hg]? NO Repair : Inspect exhaust system for source of high restriction.	Repair complete